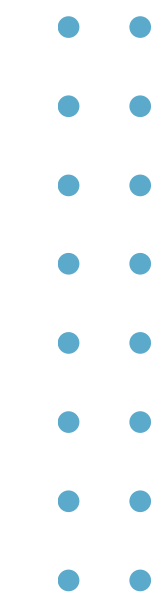


ASSIGNMENT



Build a Robot Distance Sensor Program

Create a Python program that reads a list of distance measurements from a robot's sensors and decides what the robot should do.
Use Claude to help write the code, then improve it yourself.
Push your code to GitHub.



The Task

A robot has a distance sensor on its front. The sensor takes multiple measurements and gives you a LIST of distances (in meters). Your program must read this list and for each distance, tell the robot what to do:

If distance less than 0.5m: STOP (obstacle too close)

If distance 0.5m to 1m: SLOW (obstacle nearby)

If distance more than 1m: MOVE FAST (path is clear)

Example: If the sensor gives you [0.3, 1.5, 0.8, 2.0, 0.4], your program should output: STOP, MOVE, SLOW, MOVE, STOP

Your One Python File Must Have

- ✓ Comments at top explaining what the program does
- ✓ Robot class with name and battery
- ✓ Method that processes a list of distances
- ✓ Comments explaining what each method does
- ✓ Test code at the bottom (at least 5 test cases)
- ✓ Error handling for bad distance values



Use AI to Help

Ask AI to write the code

Example: "Create a Robot class that takes a LIST of distance readings and returns actions (STOP, SLOW, MOVE)"

Read and understand it

Ask AI to explain each part. Understand the loop that processes the list.

Add comments and test code

Ask AI to add clear comments and test code at the bottom of the file.

Test and improve

Run the code. Ask AI to improve variable names and add more comments.



GitHub Repository Setup



Repository Name:

robot-distance-sensor-[YOUR-NAME]

Example: robot-distance-sensor-mohamed-ahmed

Files in Your Repo



robot.py (the code)

Robot class AND test code all in one file



README.md

Explain how your code works



README.md Must Explain:

1. What does this program do?
2. How does the Robot class work?
3. What does each method do?
4. How do I run the code?
5. What did you learn from using AI?



What to Submit



GitHub Repository Link

Submit the link to your GitHub repo. Make sure it's PUBLIC so we can see it.

Checklist Before Submitting

- Repository name is correct: robot-distance-sensor-[YOUR-NAME]
- robot.py has the Robot class and test code
- README.md explains your code and what you learned
- Repository is PUBLIC (not private)
- Code is commented and easy to read
- At least 5 test cases in robot.py

